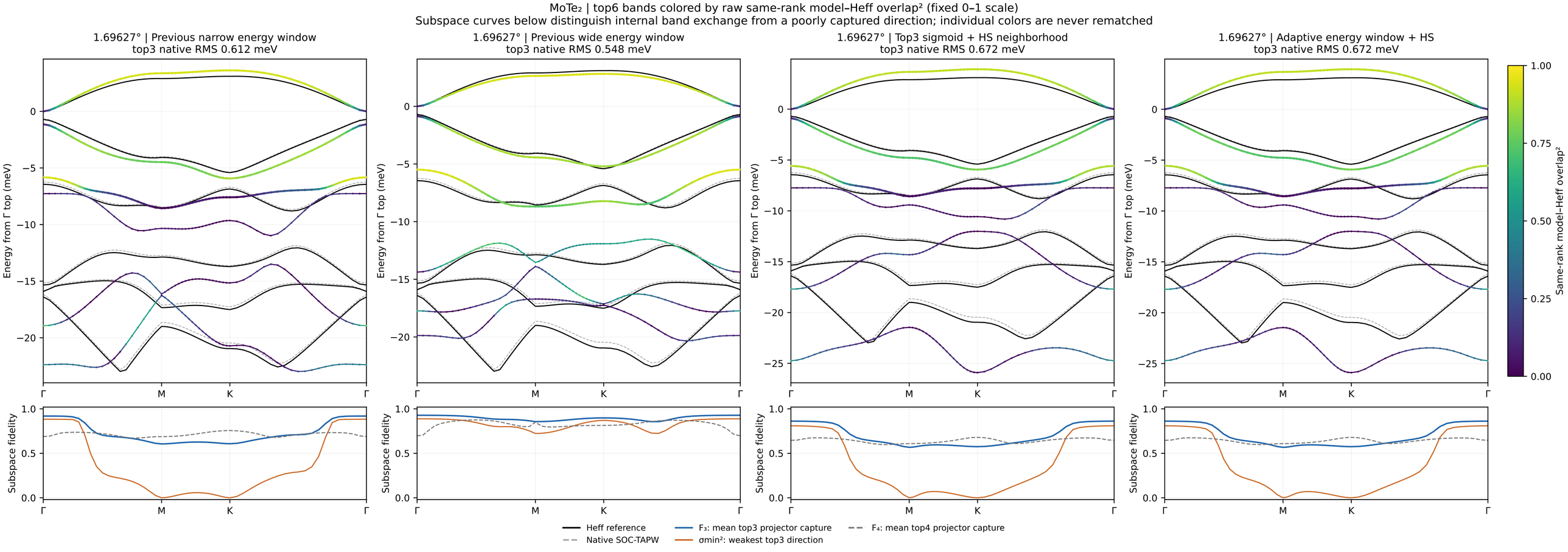
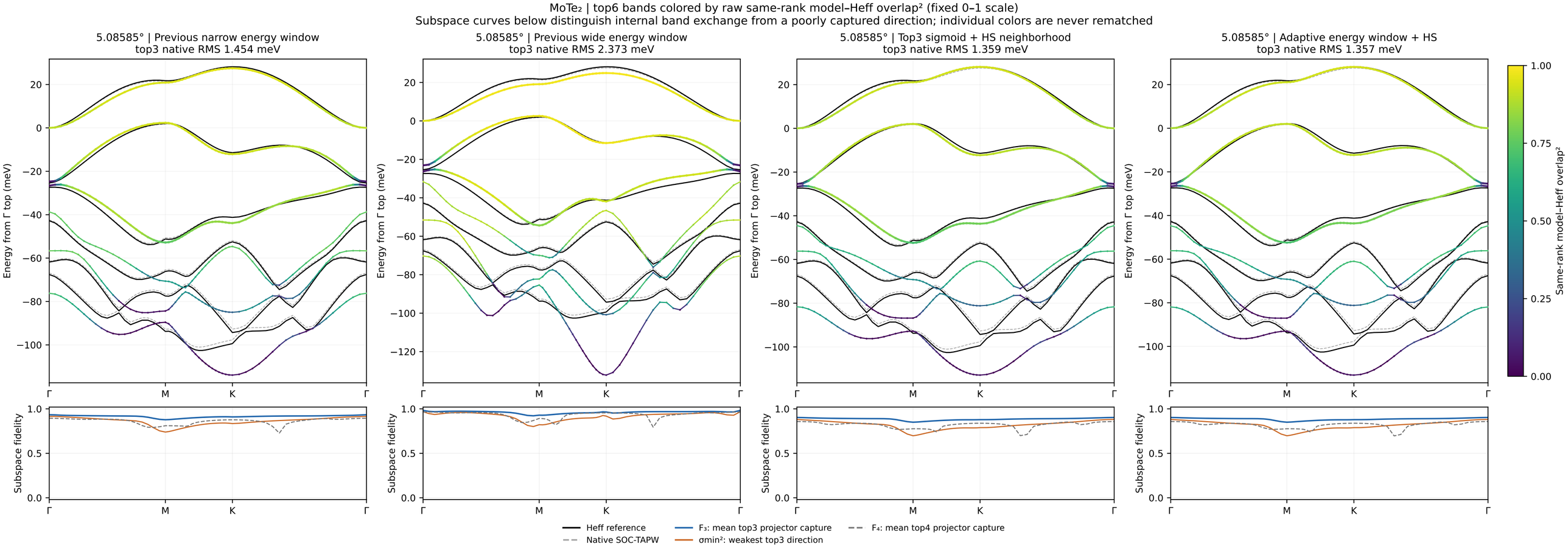
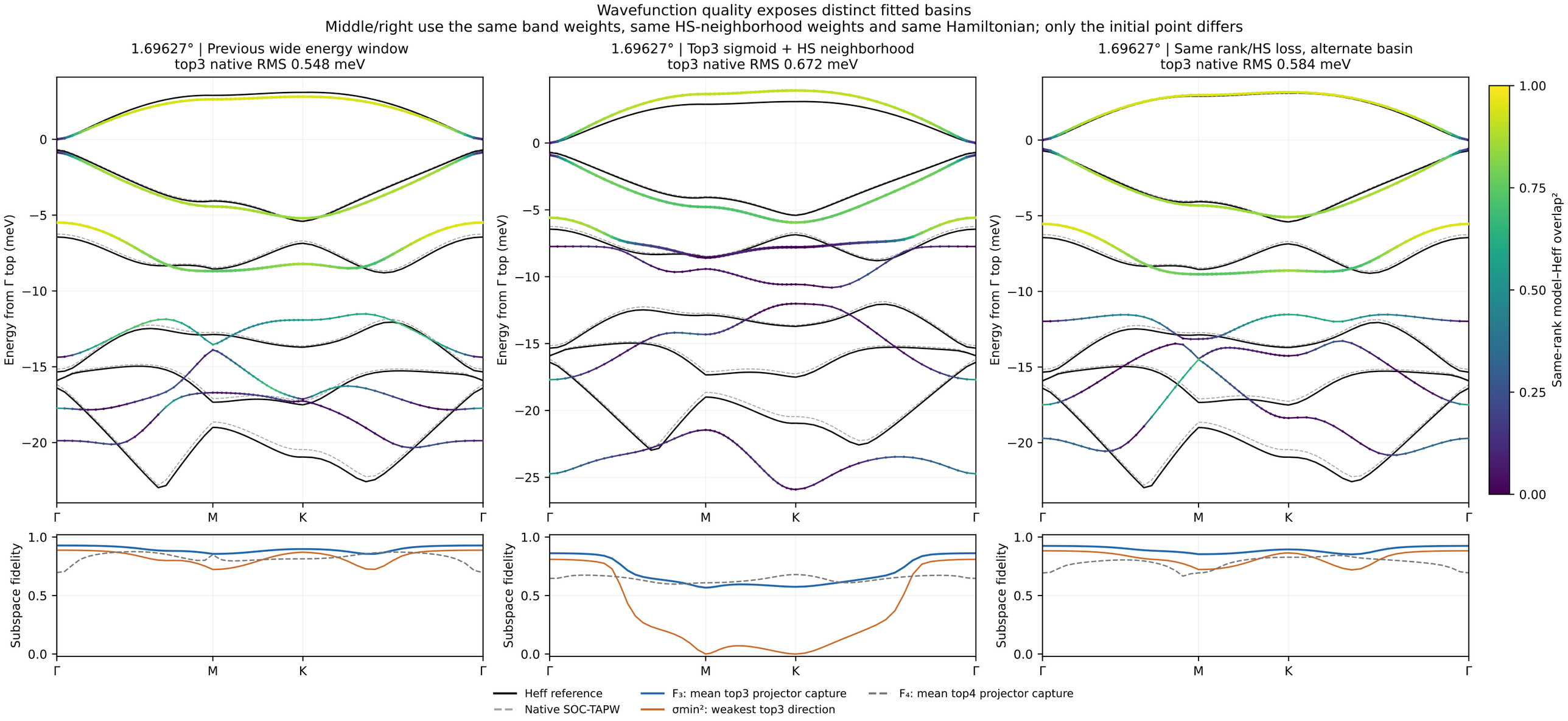
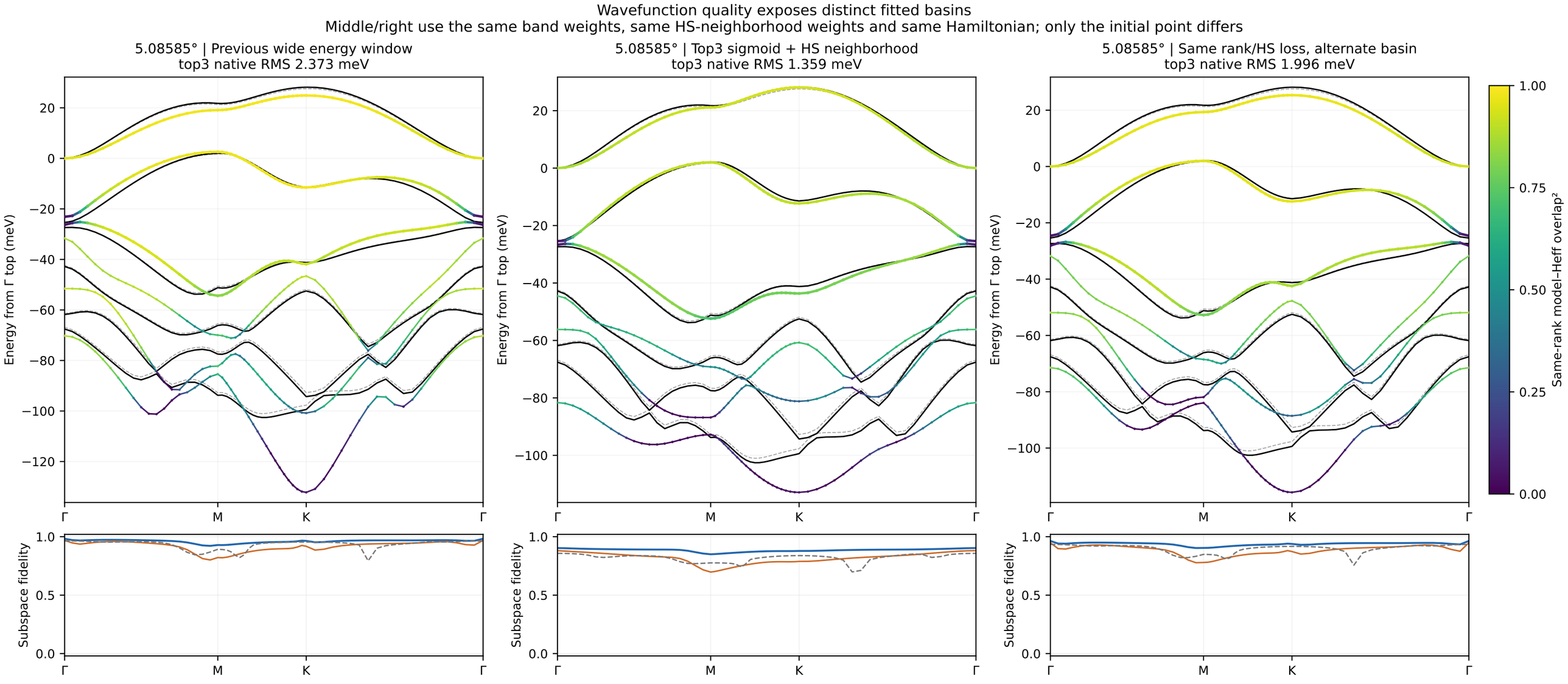

MoTe₂ | top6 bands colored by raw same-rank model–Heff overlap² (fixed 0–1 scale)
Subspace curves below distinguish internal band exchange from a poorly captured direction; individual colors are never rematched
1.69627° | Previous narrow energy window
top3 native RMS 0.612 meV
1.69627° | Previous wide energy window
top3 native RMS 0.548 meV
1.69627° | Top3 sigmoid + HS neighborhood
top3 native RMS 0.672 meV
1.69627° | Adaptive energy window + HS
top3 native RMS 0.672 meV
Energy from Γ top (meV)
Subspace fidelity
Γ
M
K
Same-rank model-Heff overlap²
Heff reference
Native SOC-TAPW
F₃: mean top3 projector capture
σmin²: weakest top3 direction
F₄: mean top4 projector capture


MoTe₂ | top6 bands colored by raw same-rank model–Heff overlap² (fixed 0–1 scale)
Subspace curves below distinguish internal band exchange from a poorly captured direction; individual colors are never rematched
5.08585° | Previous narrow energy window
top3 native RMS 1.454 meV
5.08585° | Previous wide energy window
top3 native RMS 2.373 meV
5.08585° | Top3 sigmoid + HS neighborhood
top3 native RMS 1.359 meV
5.08585° | Adaptive energy window + HS
top3 native RMS 1.357 meV
Energy from Γ top (meV)
Subspace fidelity
Γ
M
K
Same-rank model–Heff overlap²
Heff reference
Native SOC-TAPW
F₃: mean top3 projector capture
σmin²: weakest top3 direction
F₄: mean top4 projector capture


Wavefunction quality exposes distinct fitted basins
Middle/right use the same band weights, same HS-neighborhood weights and same Hamiltonian; only the initial point differs
1.69627° | Previous wide energy window
top3 native RMS 0.548 meV
1.69627° | Top3 sigmoid + HS neighborhood
top3 native RMS 0.672 meV
1.69627° | Same rank/HS loss, alternate basin
top3 native RMS 0.584 meV
Energy from Γ top (meV)
Subspace fidelity
Same-rank model–Heff overlap²
Γ
M
K
Heff reference
Native SOC-TAPW
F₃: mean top3 projector capture
σmin²: weakest top3 direction
F₄: mean top4 projector capture


Wavefunction quality exposes distinct fitted basins
Middle/right use the same band weights, same HS-neighborhood weights and same Hamiltonian; only the initial point differs
5.08585° | Previous wide energy window
top3 native RMS 2.373 meV
5.08585° | Top3 sigmoid + HS neighborhood
top3 native RMS 1.359 meV
5.08585° | Same rank/HS loss, alternate basin
top3 native RMS 1.996 meV
Energy from Γ top (meV)
Subspace fidelity
Same-rank model–Heff overlap²
Γ
M
K
Heff reference
Native SOC-TAPW
F₃: mean top3 projector capture
σmin²: weakest top3 direction
F₄: mean top4 projector capture

Previous six loss choices, now with raw model–Heff overlap² colors
Model coefficients unchanged; common 0–1 color scale; top6 shown

1.69627° | Previous hard top3 / absolute
top3 native RMS 0.672 meV

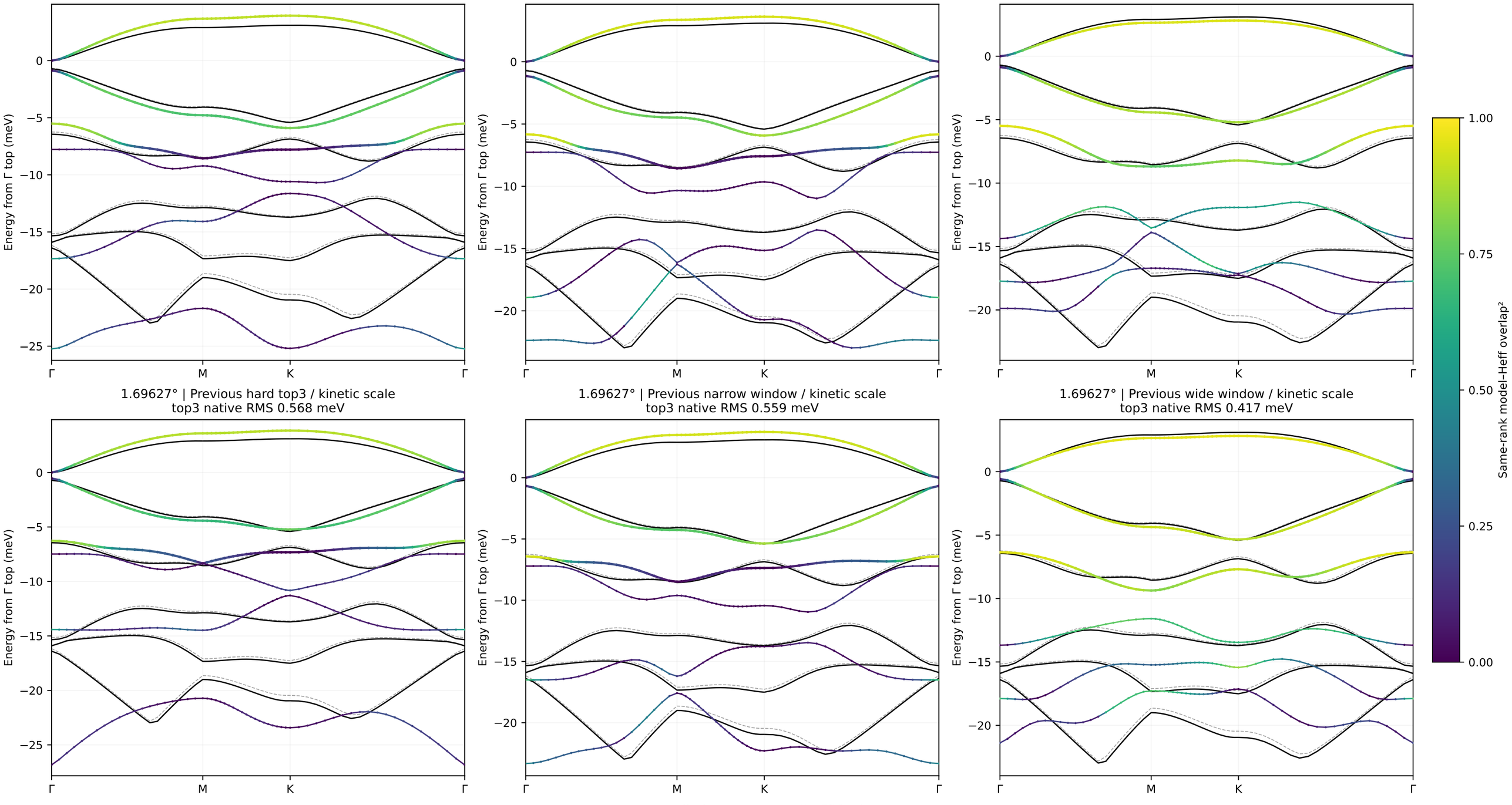
1.69627° | Previous narrow energy window
top3 native RMS 0.612 meV

1.69627° | Previous wide energy window
top3 native RMS 0.548 meV

1.69627° | Previous hard top3 / kinetic scale
top3 native RMS 0.568 meV

1.69627° | Previous narrow window / kinetic scale
top3 native RMS 0.559 meV

1.69627° | Previous wide window / kinetic scale
top3 native RMS 0.417 meV



Previous six loss choices, now with raw model–Heff overlap² colors
Model coefficients unchanged; common 0–1 color scale; top6 shown

5.08585° | Previous hard top3 / absolute
top3 native RMS 1.398 meV

5.08585° | Previous narrow energy window
top3 native RMS 1.454 meV

5.08585° | Previous wide energy window
top3 native RMS 2.373 meV

5.08585° | Previous hard top3 / kinetic scale
top3 native RMS 1.798 meV

5.08585° | Previous narrow window / kinetic scale
top3 native RMS 1.699 meV

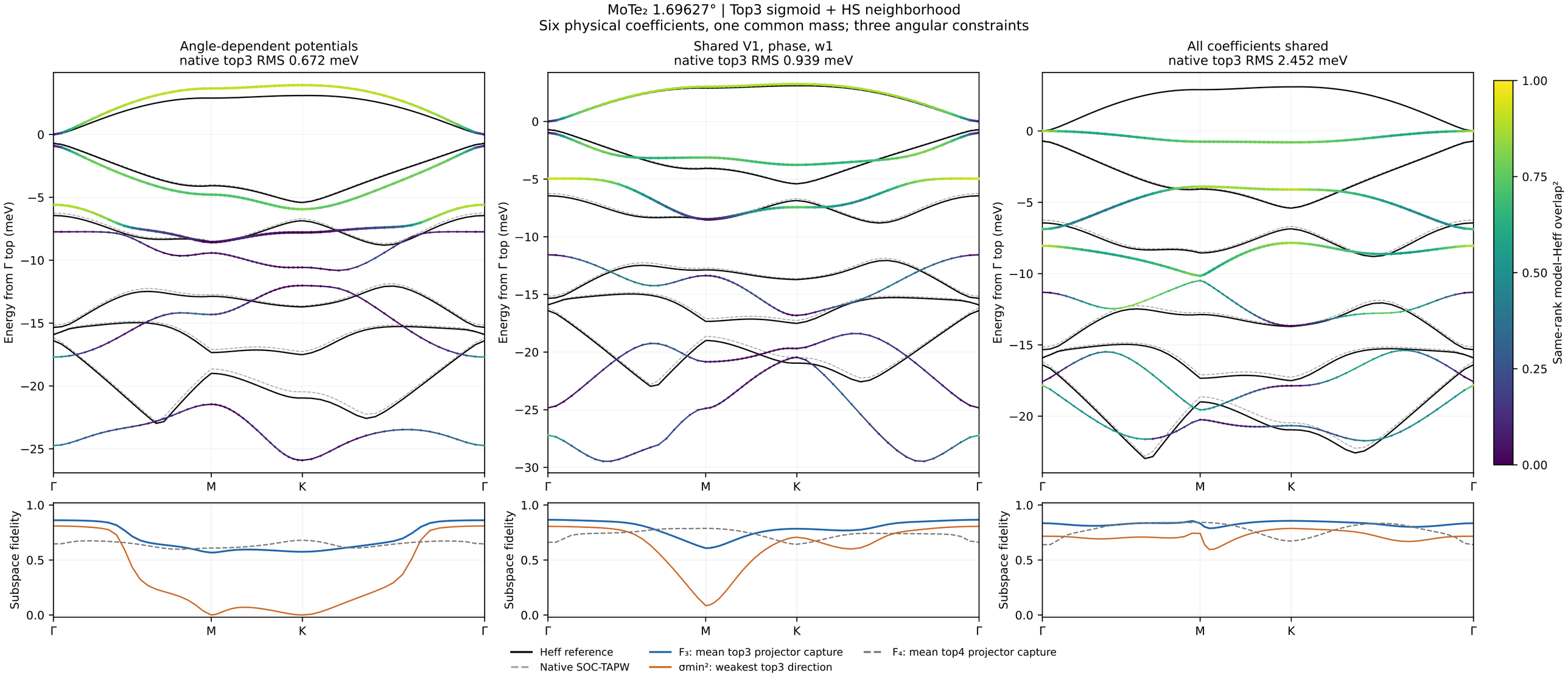
5.08585° | Previous wide window / kinetic scale
top3 native RMS 2.749 meV

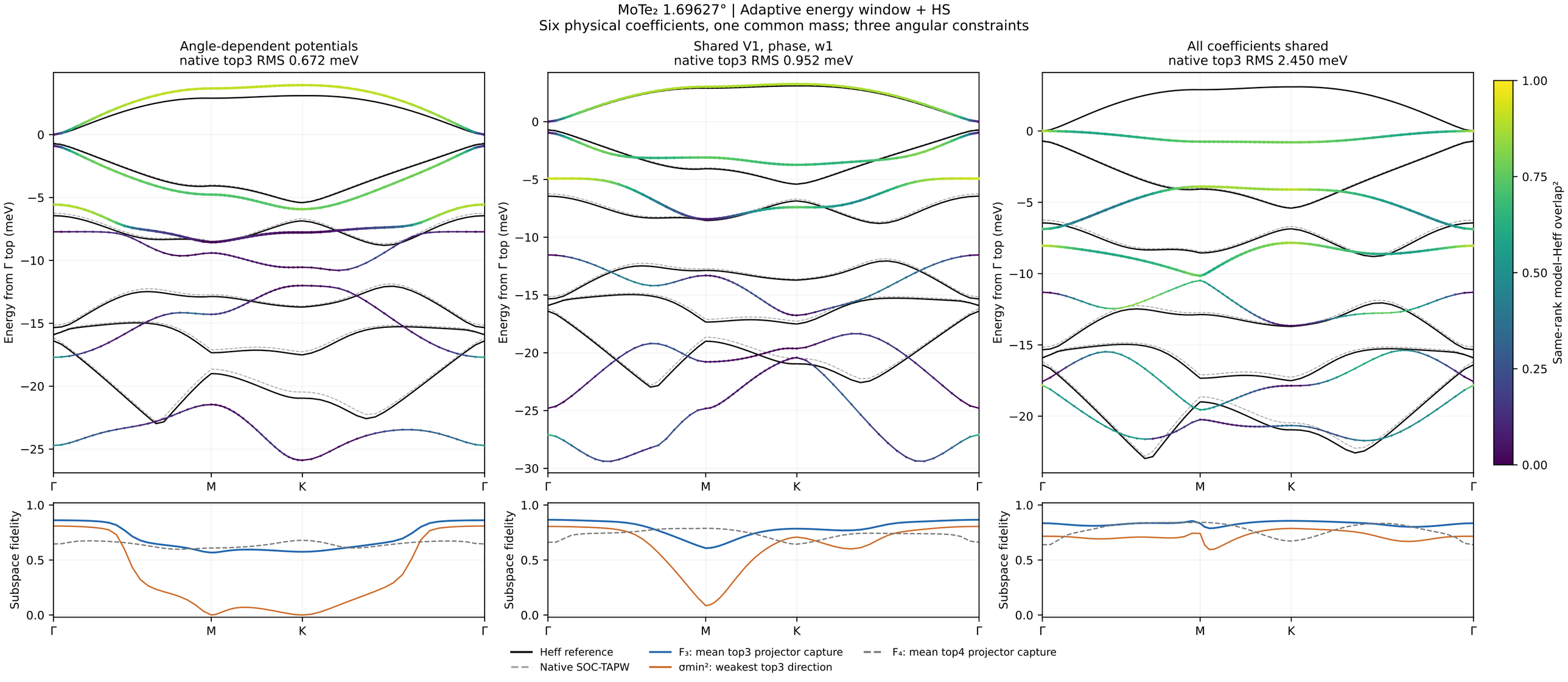
Energy from Γ top (meV)

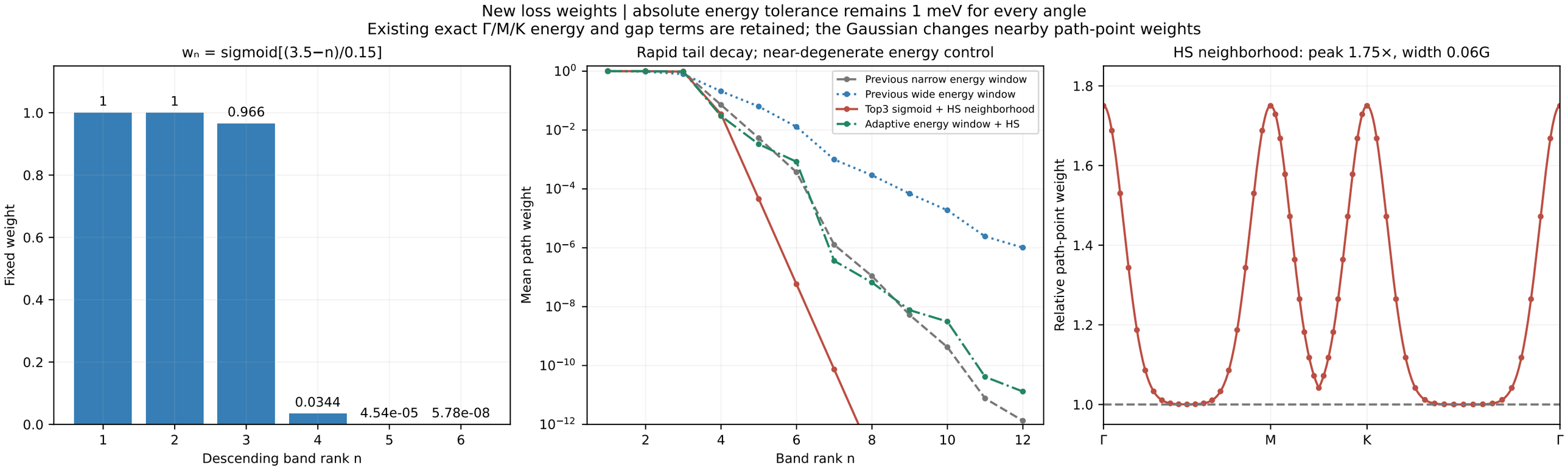
Γ M K Γ

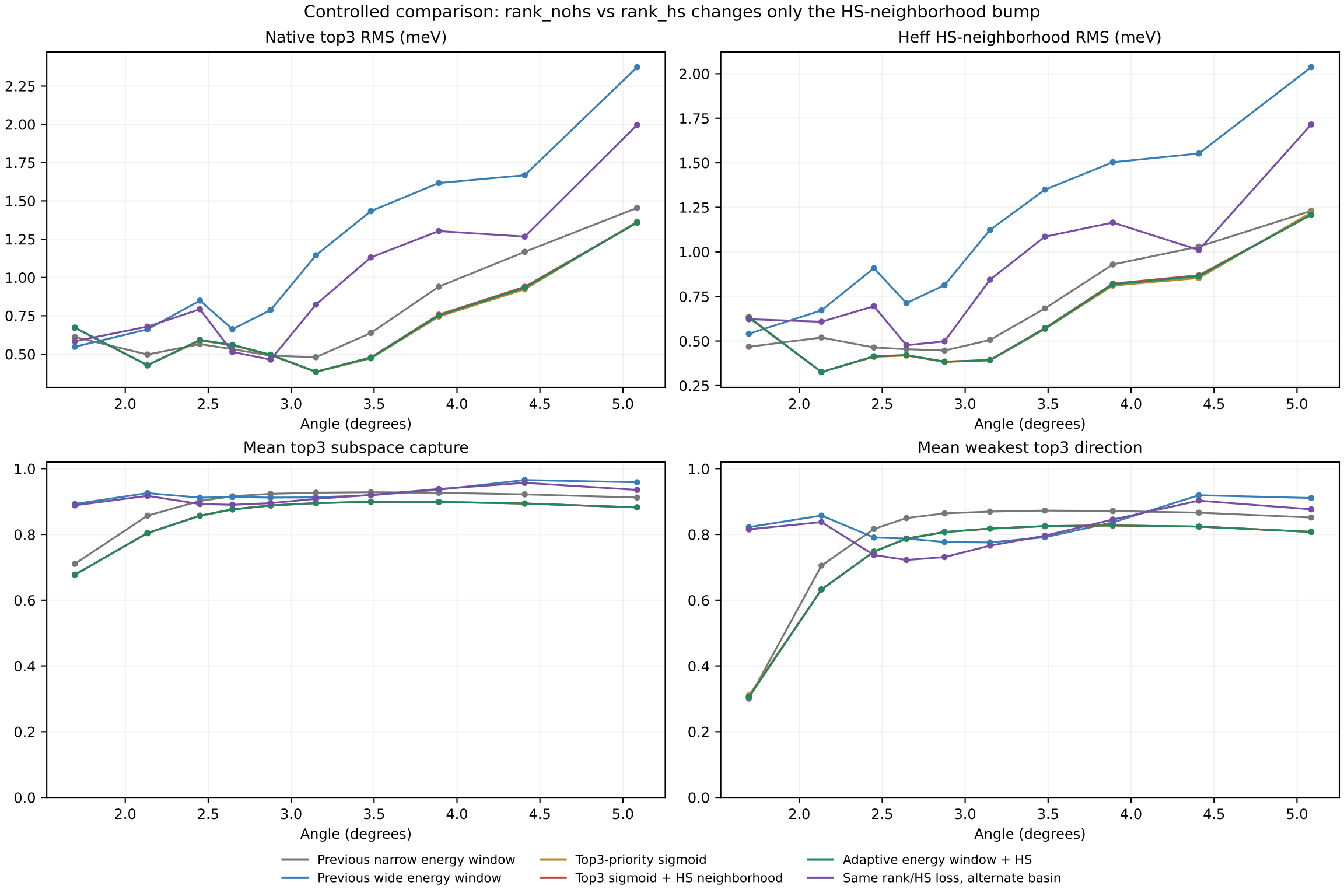
Heff reference — Native SOC-TAPW

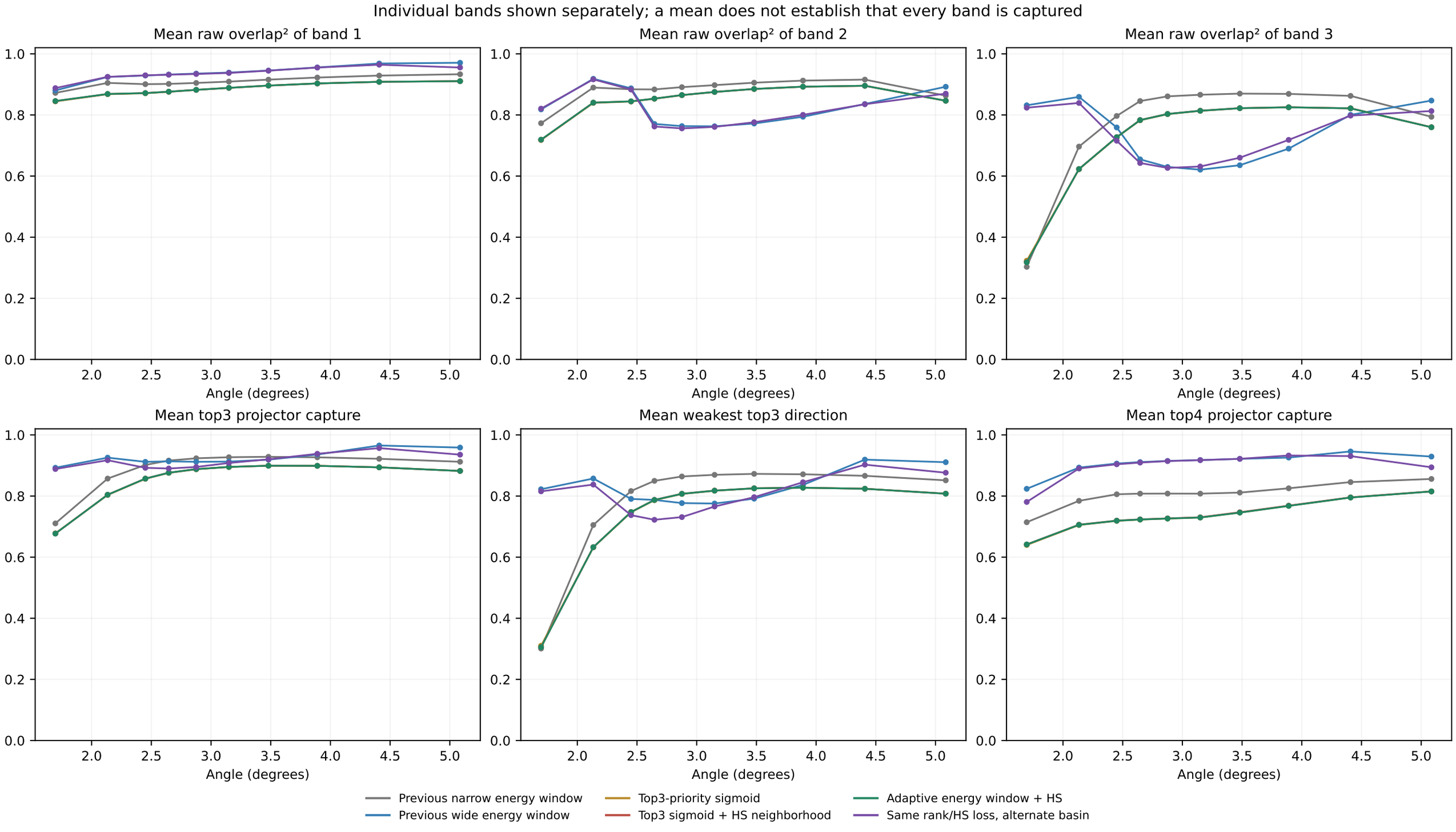
Same-rank model–Heff overlap²


MoTe₂ 1.69627° | Top3 sigmoid + HS neighborhood
Six physical coefficients, one common mass; three angular constraints
Angle-dependent potentials
native top3 RMS 0.672 meV
Shared V1, phase, w1
native top3 RMS 0.939 meV
All coefficients shared
native top3 RMS 2.452 meV
Energy from Γ top (meV)
Subspace fidelity
Same-rank model–Heff overlap²
Γ
M
K
Heff reference
Native SOC-TAPW
F₃: mean top3 projector capture
σmin²: weakest top3 direction
F₄: mean top4 projector capture


MoTe₂ 1.69627° | Adaptive energy window + HS
Six physical coefficients, one common mass; three angular constraints
Angle-dependent potentials
native top3 RMS 0.672 meV
Shared V1, phase, w1
native top3 RMS 0.952 meV
All coefficients shared
native top3 RMS 2.450 meV
Energy from Γ top (meV)
Subspace fidelity
Same-rank model–Heff overlap²
Γ
M
K
Heff reference
Native SOC-TAPW
F₃: mean top3 projector capture
σmin²: weakest top3 direction
F₄: mean top4 projector capture


New loss weights | absolute energy tolerance remains 1 meV for every angle
Existing exact Γ/M/K energy and gap terms are retained; the Gaussian changes nearby path-point weights
wₙ = sigmoid[(3.5−n)/0.15]
Fixed weight
Descending band rank n
1
1
0.966
0.0344
4.54e-05
5.78e-08
Rapid tail decay; near-degenerate energy control
Mean path weight
Band rank n
Previous narrow energy window
Previous wide energy window
Top3 sigmoid + HS neighborhood
Adaptive energy window + HS
HS neighborhood: peak 1.75×, width 0.06G
Relative path-point weight
Γ
M
K
Γ


Controlled comparison: rank_nohs vs rank_hs changes only the HS-neighborhood bump
Native top3 RMS (meV)
Heff HS-neighborhood RMS (meV)
Mean top3 subspace capture
Mean weakest top3 direction
Angle (degrees)
Previous narrow energy window
Previous wide energy window
Top3-priority sigmoid
Top3 sigmoid + HS neighborhood
Adaptive energy window + HS
Same rank/HS loss, alternate basin


Individual bands shown separately; a mean does not establish that every band is captured
Mean raw overlap² of band 1
Mean raw overlap² of band 2
Mean raw overlap² of band 3
Mean top3 projector capture
Mean weakest top3 direction
Mean top4 projector capture
Angle (degrees)
Previous narrow energy window
Previous wide energy window
Top3-priority sigmoid
Top3 sigmoid + HS neighborhood
Adaptive energy window + HS
Same rank/HS loss, alternate basin

2.45°: distinguish near-degenerate basis rotations from actual mismatch
Groups connect adjacent model OR reference gaps below 0.2 meV; no band permutation or overlap renormalization

Raw same-rank overlap²
Previous wide energy window

Raw same-rank overlap²
Top3 sigmoid + HS neighborhood

Raw same-rank overlap²
Same rank/HS loss, alternate basin

Energy from Γ top (meV)

Near-degenerate group fidelity
Previous wide energy window

Near-degenerate group fidelity
Top3 sigmoid + HS neighborhood

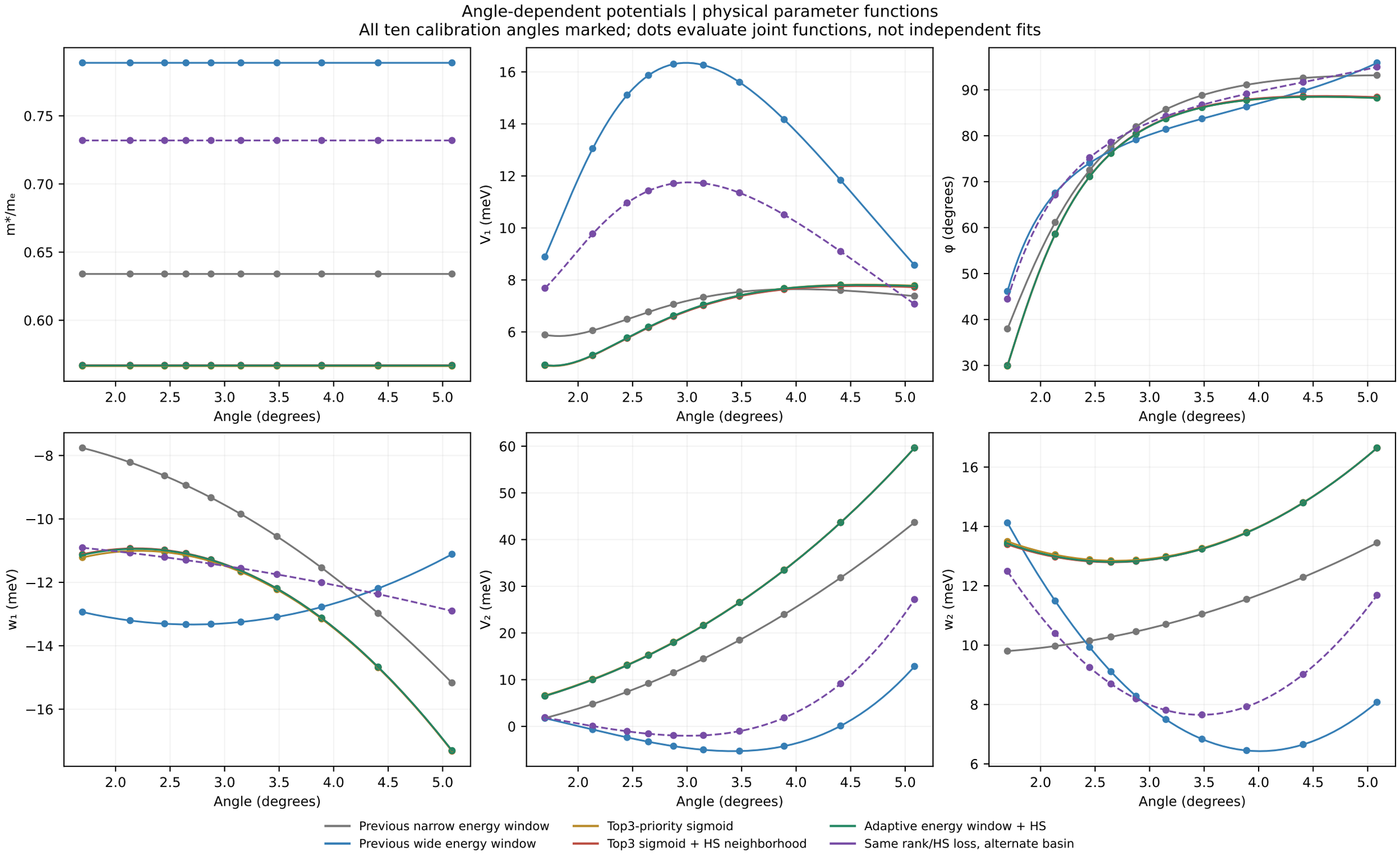
Near-degenerate group fidelity
Same rank/HS loss, alternate basin

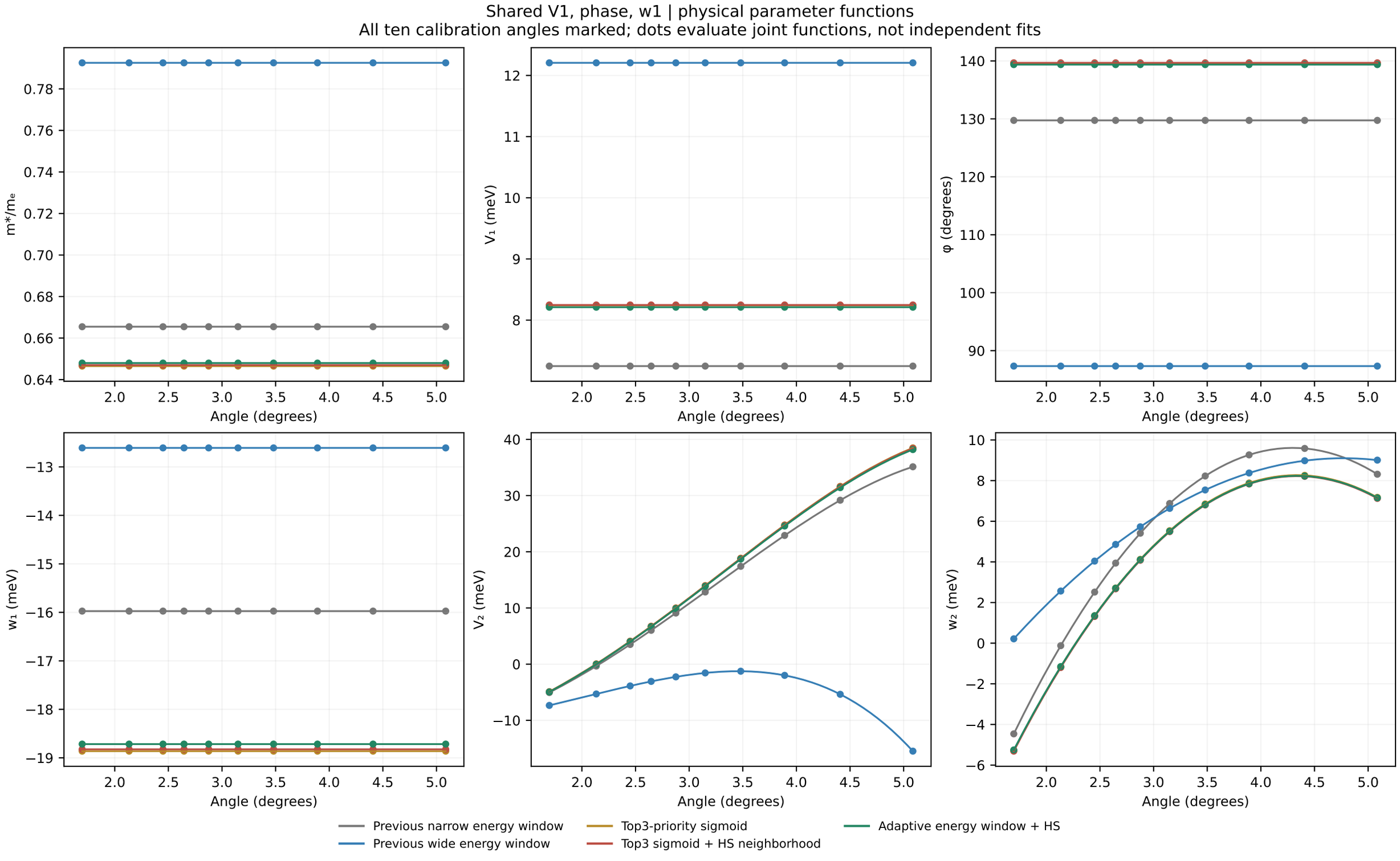
Γ M K Γ

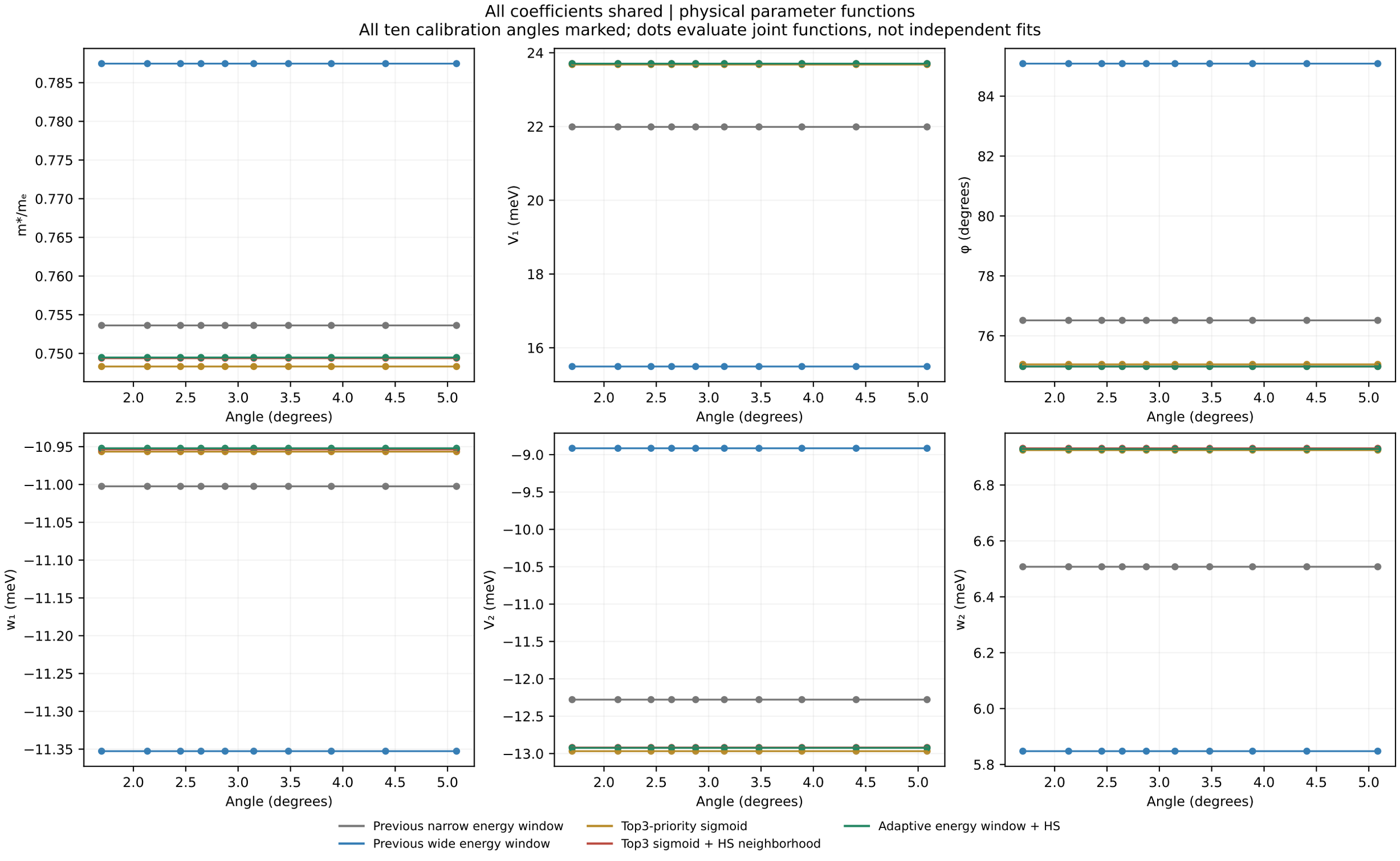
Fidelity: individual in upper row; group in lower row

Heff reference

Native SOC-TAPW


Angle-dependent potentials | physical parameter functions
All ten calibration angles marked; dots evaluate joint functions, not independent fits
m*/mₑ
V₁ (meV)
φ (degrees)
w₁ (meV)
V₂ (meV)
w₂ (meV)
Angle (degrees)
Previous narrow energy window
Previous wide energy window
Top3-priority sigmoid
Top3 sigmoid + HS neighborhood
Adaptive energy window + HS
Same rank/HS loss, alternate basin


Shared V1, phase, w1 | physical parameter functions
All ten calibration angles marked; dots evaluate joint functions, not independent fits
m*/mₑ
V₁ (meV)
φ (degrees)
w₁ (meV)
V₂ (meV)
w₂ (meV)
Angle (degrees)
Previous narrow energy window
Previous wide energy window
Top3-priority sigmoid
Top3 sigmoid + HS neighborhood
Adaptive energy window + HS


All coefficients shared | physical parameter functions
All ten calibration angles marked; dots evaluate joint functions, not independent fits
m*/mₑ
V₁ (meV)
φ (degrees)
w₁ (meV)
V₂ (meV)
w₂ (meV)
Angle (degrees)
Previous narrow energy window
Previous wide energy window
Top3-priority sigmoid
Top3 sigmoid + HS neighborhood
Adaptive energy window + HS

All angles | top6 model bands colored by raw model–Heff overlap²
Same 0–1 color scale; no rematching, no per-panel color rescaling

1.69627° | Previous wide energy window
top3 native RMS 0.548 meV

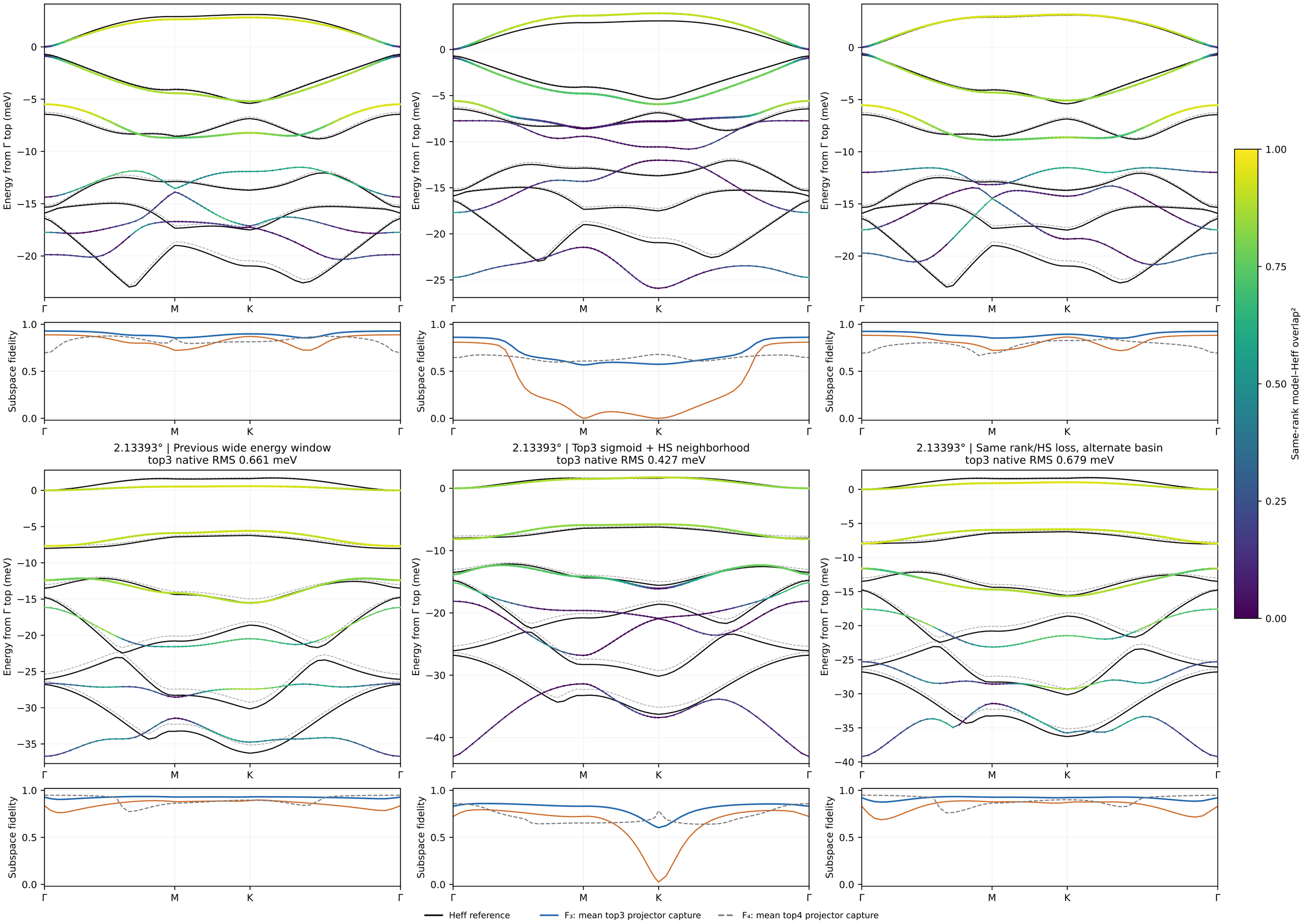
1.69627° | Top3 sigmoid + HS neighborhood
top3 native RMS 0.672 meV

1.69627° | Same rank/HS loss, alternate basin
top3 native RMS 0.584 meV

2.13393° | Previous wide energy window
top3 native RMS 0.661 meV

2.13393° | Top3 sigmoid + HS neighborhood
top3 native RMS 0.427 meV

2.13393° | Same rank/HS loss, alternate basin
top3 native RMS 0.679 meV



All angles | top6 model bands colored by raw model–Heff overlap²
Same 0–1 color scale; no rematching, no per-panel color rescaling

2.44998° | Previous wide energy window
top3 native RMS 0.849 meV

2.44998° | Top3 sigmoid + HS neighborhood
top3 native RMS 0.591 meV

2.44998° | Same rank/HS loss, alternate basin
top3 native RMS 0.792 meV

2.64591° | Previous wide energy window
top3 native RMS 0.663 meV

2.64591° | Top3 sigmoid + HS neighborhood
top3 native RMS 0.559 meV

2.64591° | Same rank/HS loss, alternate basin
top3 native RMS 0.514 meV

Energy from Γ top (meV)
Subspace fidelity
Γ M K Γ
Same-rank model–Heff overlap²
Heff reference
Native SOC-TAPW
F₃: mean top3 projector capture
σmin²: weakest top3 direction
F₄: mean top4 projector capture

All angles | top6 model bands colored by raw model–Heff overlap²
Same 0–1 color scale; no rematching, no per-panel color rescaling

2.87589° | Previous wide energy window
top3 native RMS 0.787 meV

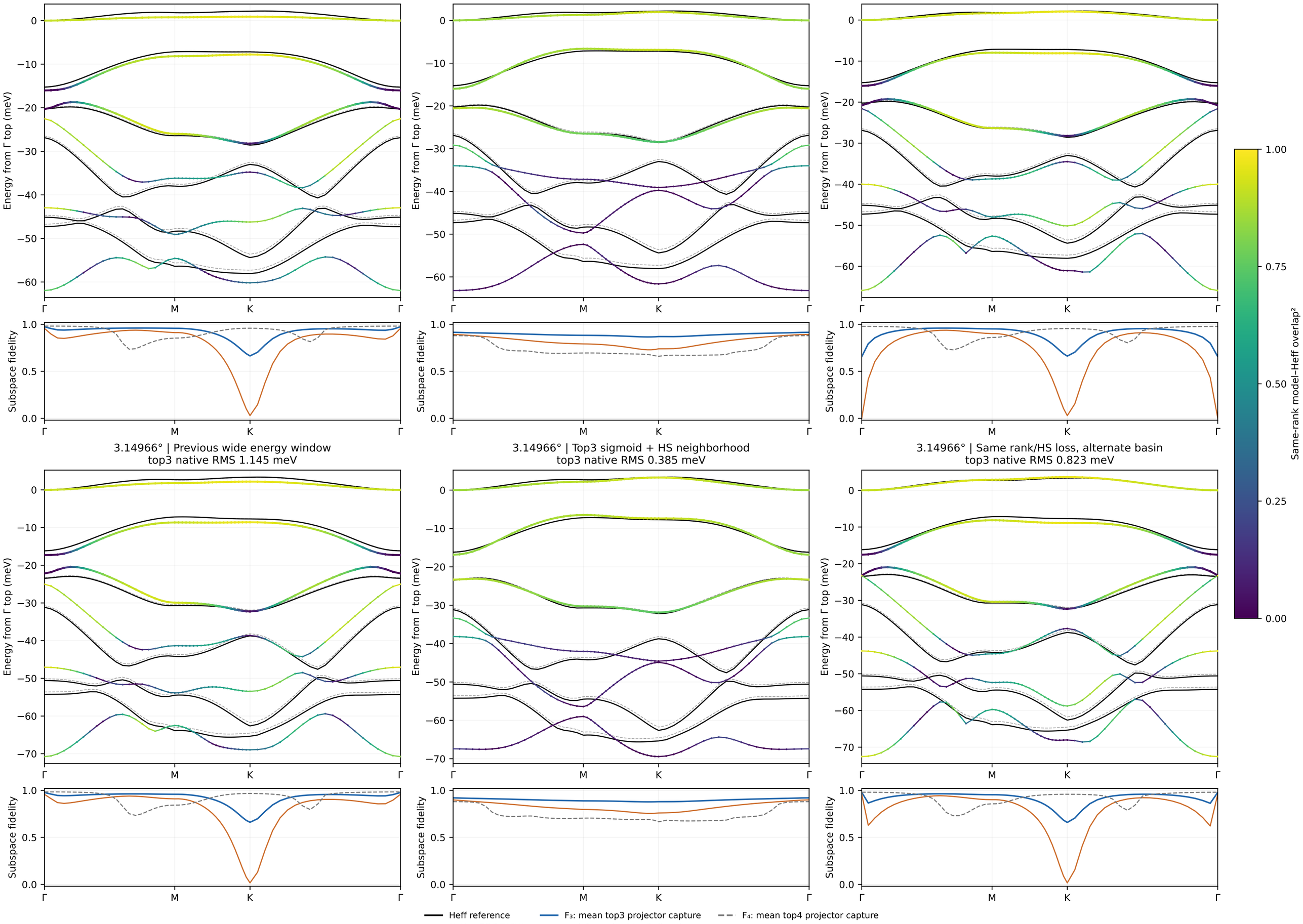
2.87589° | Top3 sigmoid + HS neighborhood
top3 native RMS 0.494 meV

2.87589° | Same rank/HS loss, alternate basin
top3 native RMS 0.464 meV

3.14966° | Previous wide energy window
top3 native RMS 1.145 meV

3.14966° | Top3 sigmoid + HS neighborhood
top3 native RMS 0.385 meV

3.14966° | Same rank/HS loss, alternate basin
top3 native RMS 0.823 meV



All angles | top6 model bands colored by raw model–Heff overlap²
Same 0–1 color scale; no rematching, no per-panel color rescaling

3.48101° | Previous wide energy window
top3 native RMS 1.433 meV

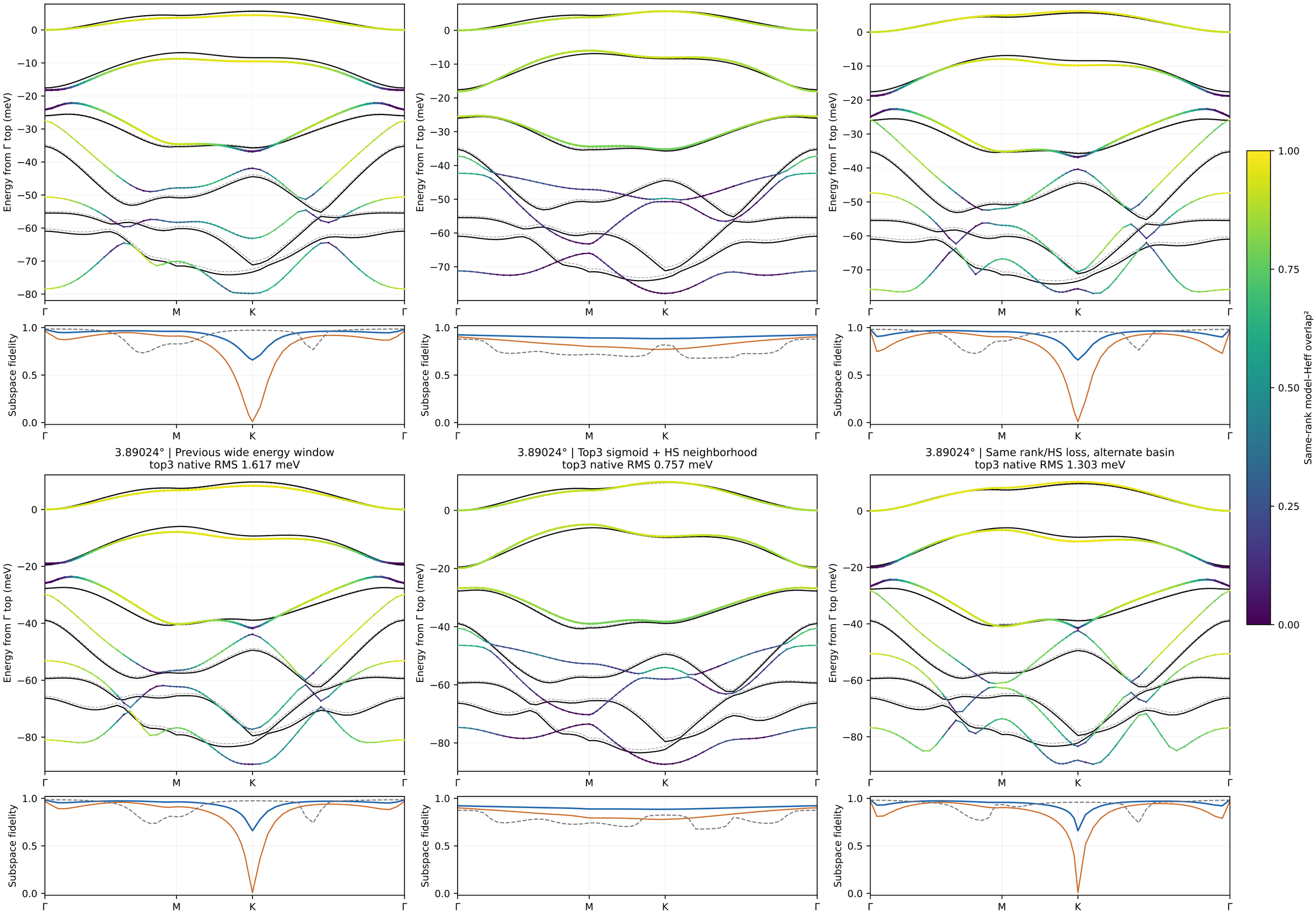
3.48101° | Top3 sigmoid + HS neighborhood
top3 native RMS 0.479 meV

3.48101° | Same rank/HS loss, alternate basin
top3 native RMS 1.131 meV

3.89024° | Previous wide energy window
top3 native RMS 1.617 meV

3.89024° | Top3 sigmoid + HS neighborhood
top3 native RMS 0.757 meV

3.89024° | Same rank/HS loss, alternate basin
top3 native RMS 1.303 meV



All angles | top6 model bands colored by raw model–Heff overlap²
Same 0–1 color scale; no rematching, no per-panel color rescaling

4.40846° | Previous wide energy window
top3 native RMS 1.667 meV

4.40846° | Top3 sigmoid + HS neighborhood
top3 native RMS 0.939 meV

4.40846° | Same rank/HS loss, alternate basin
top3 native RMS 1.267 meV

5.08585° | Previous wide energy window
top3 native RMS 2.373 meV

5.08585° | Top3 sigmoid + HS neighborhood
top3 native RMS 1.359 meV

5.08585° | Same rank/HS loss, alternate basin
top3 native RMS 1.996 meV

